A. Mill Equipment

1. 510mm Heavy Duty Roughing Mill

1.1 High Speed Flywheel **1 No**

Steel Flywheel running @ 740 rpm
app. 8 Tons weight complete with shaft,
spherical roller bearings, pedestals and base plate.

1.2 Reduction Gear Box. **2 Nos.**

Input - 740 rpm
Output - 95 rpm
Nominal Power - 1120 KW
Peak load - 2240 KW
Momentary Load - 3360 KW

Construction:

- Gears & Pinions will be of forged Steel, shrunk fitted on En-9 forged steel shafts.
- Helical teeth Profile on Gear Hobbing Machine
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.

Transparent window provided on both sides of the housing to observe teeth and lubrication

1.3 510mm 3 -Hi Pinion Gear Box. **1 No.**

Pinion Centre - 510 mm
Nominal Power - 1120 KW
Peak load - 2240 KW
Momentary Load - 3360 KW

Construction:

- Gears & Pinions will be of forged Steel, shrunk fitted on En-9 forged steel shafts.
- Helical teeth Profile on Gear Hobbing Machine
- Shafts end with key way.
- Pinion shaft mounted on anti-friction bearings.
- Mechanical Seals will be provided to avoid oil Leakage.
- Piping provided for forced lubrication arrangement of bearings and pinion teeth.
- Oil seal provided to stop oil leakage and ingress of water into the housing.
- Pinion enclosed in Steel Fabricated housing, duly stress relieved & machined to close tolerance.

Transparent window provided on both sides of the housing to observe teeth and lubrication

1.4 Base Plate for Pinion Gear Box **2 Nos.**

1.5 Mill Stand **1 No.**

Type : 3 Hi
Size : 510MM
Roll Dia. : Max. 550 MM
Min. 490 MM
Barrel : 1350 MM
Roll Neck Bearing : Spherical Roller Bearing
Housing & Cap : Cast Steel – Grade-2 (IS-1030)
All Fasteners : High Tensile Steel Gr.8.8 min.
Material for Chocks : Cast Steel with 500-600 N/mm2 U.T.S.
Chocks are accurately machined and are inter changeable.
Top Roll Adjustment : By screw down mechanism.

- The housing shall be accurately machined on CNC machine and fitted with wear plates by high tensile Allen Cap Screw. This ensures accurate fitting of chocks in housings. The wear plates are ground Finished
- The cap shall be fixed over the housings by means of two tapper wedges which gives very good rigidity and does not loose during Mill operation.
- Mill Stand will be complete with all fitments, with Cooling Water piping and valves, roll cooling and Guide Cooling and Foundation bolts etc.
- The accurate machining of parts and housings keep vibration level to minimum during rolling.

1.6 Cardan Shaft & Universal Head **3 Nos.**

Location : Between Pinion Stand and rolls
Type : Cross Joint Type
Material of coupling heads : EN-9 (Forged)

CONSTRUCTION

- Cardan Shaft & Couplings are be required to connect output shafts of Pinion Stand to rolls of mill stand.
- Each set shall consist of two couplings hubs assemblies and one cardan Shaft.
- Cardan Shaft will be selected for the required duties from renowned manufacturers.
- The connecting Flange shall have keys to ensure positive transmission and reduce the loads on bolts.
- Connecting Hub suitable for roll end on one side & pinion end on the other side are provided.

1.7 Full Geared Heavy Duty Couplings **4 Set**

Type	:	full flexible Geared Coupling to be connected between Reduction Gear Box, Pinion Stand and Motor
Design Parameters		
Normal Rating	:	1120 kW
RPM	:	750
Service Factor	:	≥ 3
Type of Drive	:	Non reversible for all
Duty	:	Rolling Mill Duty

1.8 Foundation Bolts **1 Set**

1.9 Rolls (1 Set) **3 Nos.**

1.10 Roll Neck Bearing (1 Set) **6 Nos.**

1.11 Motor Base Plate **1 Set**